Profiling_Lazy_Optimization.json

Server feedback (status/metrics/errors):


```
{
 "task_id": "parallel_task_000336_b4243221",
 "status": "completed",
 "compiled": true,
 "correctness": true,
 ”hacking": false,
 "reference_runtime": 47.4,
 "kernel_runtime": 47.0,
 "speedup": 1.01,
 "metadata": {
  …
  "hardware": "NVIDIA H100 80GB HBM3",
  "profiling": {
   "kernels": [
    {
     "name": "void convolveNd_dgrad_float_engine<float, 3…",
     "cuda_time_us": 27522.25,
     "cpu_time_us": 0.0,
     "count": 5
    },
    {
     "name": "void cudnn::engines_precompiled::setTensor5d…",
     …
    },
    …
    {
     "name": "sum_channels_kernel",
     "cuda_time_us": 34.8,
     "cpu_time_us": 0.0,
     "count": 5
    }
   ],
   "total_cuda_time_us": 237500.34,
   "total_cpu_time_us": 479674.12,
   "memory_stats": {
    ...
   }
  },
  …
  "triton_profiler_matches": "[…,'sum_channels_kernel grid…']",
  "custom_kernel_cuda_time_in_profiling_us": 34.82,
   "total_kernel_run_time_in_profiling_us": 237500.34,
   "custom_kernel_cuda_time_coverage": "Custom kernel CUDA time: 34.82us / Total time:
237500.35us, Coverage: 0.014%",
  …
```


Profiling_Better_Fusion.json

Server feedback (status/metrics/errors):


```
{
 …
 "reference_runtime": 47.5,
 "kernel_runtime": 22.8,
 "speedup": 2.08,
 "metadata": {
  "hardware": "NVIDIA H100 80GB HBM3",
  "profiling": {
   "kernels": [
    {
     "name": "max_pool3d_k2s2_per_channel_kernel",
     "cuda_time_us": 96169.16,
     "cpu_time_us": 0.0,
     "count": 5
    },
    {
     "name": "sm90_xmma_dgrad_implicit_gemm_...",
     "cuda_time_us": 6663.94,
     "cpu_time_us": 0.0,
     "count": 5
    },
    {
     "name": "max_pool3d_k3s3_per_channel_kernel",
     …
    },
    …
    {
     "name": "sum_channels_kernel",
     …
    ],
   "total_cuda_time_us": 118674.11,
   "total_cpu_time_us": 243259.05,
   "kernel_count": 8,
   "memory_stats": {
    …
   }
  },
  …
  "triton_profiler_matches": "[…'max_pool3d_k2s2_per_channel_kernel grid…’,
              'max_pool3d_k3s3_per_channel_kernel…', 'sum_channels_kernel…’]”,
  "custom_kernel_cuda_time_in_profiling_us": 102242.61,
  "total_kernel_run_time_in_profiling_us": 118674.10,
  "custom_kernel_cuda_time_coverage": "Custom kernel CUDA time: 102242.61us / Total time:
118674.11us, Coverage: 86.15%",
  …
```